

ASSIGNMENT - 8

Q1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
KD3-98843-Nandini> SELECT * FROM orders WHERE cnum IN ( SELECT cnum FROM customers WHERE cname = 'Cisneros');
```

onum	amt	odate	cnum	snum
3001	18.69	1990-10-03	2008	1007
3006	1098.16	1990-10-03	2008	1007

```
2 rows in set (0.00 sec)
```

Q7. Extract all the orders of Motika.

```
KD3-98843-Nandini> SELECT * FROM orders WHERE snum = (SELECT snum FROM salespeople WHERE sname = 'Motika');
```

onum	amt	odate	cnum	snum
3002	1900.10	1990-10-03	2007	1004

```
1 row in set (0.00 sec)
```

Q8. Find all the orders of salespeople servicing customers in London.

```
KD3-98843-Nandini> SELECT * FROM orders WHERE snum IN (SELECT snum FROM salespeople WHERE city = 'London');
```

onum	amt	odate	cnum	snum
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3008	4723.00	1990-10-04	2006	1001
3011	9891.88	1990-10-04	2006	1001

```
4 rows in set (0.01 sec)
```

Q6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
KD3-98843-Nandini> SELECT * FROM orders WHERE amt > ANY (
->     SELECT amt FROM orders
->     INNER JOIN customers ON orders.cnum = customers.cnum WHERE
customers.city = 'London'
-> );
```

onum	amt	odate	cnum	snum
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3008	4723.00	1990-10-04	2006	1001
3011	9891.88	1990-10-04	2006	1001

6 rows in set (0.01 sec)

Q4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
KD3-98843-Nandini> SELECT * FROM customers
-> WHERE rating >= ANY (SELECT rating FROM customers
->     INNER JOIN salespeople ON customers.snum = salespeople.snum
WHERE salespeople.sname = 'Serres'
-> );
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.00 sec)

Q11. Select customers who have a greater rating than any other customer in Rome.

```
KD3-98843-Nandini> SELECT * FROM customers WHERE rating > ANY
-> (
->     SELECT rating FROM customers WHERE city = 'Rome'
-> );
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.00 sec)

Q12. Select all orders that had amounts that were greater than at least one of the orders from '1990-10-04'.

```
KD3-98843-Nandini> SELECT * FROM orders WHERE amt > ANY
-> (
->   SELECT amt FROM orders WHERE odate = '1990-10-04'
-> );
```

onum	amt	odate	cnum	snum
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002
3011	9891.88	1990-10-04	2006	1001

8 rows in set (0.01 sec)

Q13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
KD3-98843-Nandini> SELECT * FROM orders WHERE amt < ANY
-> (
->   SELECT O.amt FROM orders O
->   INNER JOIN customers C ON O.cnum = C.cnum
->   WHERE C.city = 'San Jose'
-> );
```

onum	amt	odate	cnum	snum
3001	18.69	1990-10-03	2008	1007
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3007	75.75	1990-10-04	2004	1002
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002

8 rows in set (0.00 sec)

Q 2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
KD3-98843-Nandini> SELECT orders.amt, customers.cname, customers.rating
FROM orders INNER JOIN customers ON orders.cnum = customers.cnum
  WHERE orders.amt >
  (
    SELECT AVG(amt)
    FROM orders
  );
```

amt	cname	rating
5160.45	Liu	200
9891.88	Clemens	100
4723.00	Clemens	100

3 rows in set (0.01 sec)

Q 14. Select those customers whose rating are higher than every customer in London.

```
KD3-98843-Nandini> SELECT * FROM customers WHERE rating > ALL
  (
    SELECT rating FROM customers WHERE city = 'London'
  );
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.01 sec)

5. Write a query that will find all salespeople who have no customers located in their city.
(Hint: Choose the correct keyword ANY/ALL).

```
KD3-98843-Nandini> SELECT * FROM salespeople WHERE NOT EXISTS
  (
    SELECT * FROM customers WHERE customers.snum = salespeople.snum
    AND customers.city = salespeople.city
  );
```

snum	sname	city	comm
1004	Motika	London	0.11
1007	Rifkin	Barcelona	0.15
1003	Axelrod	New York	0.10

3 rows in set (0.00 sec)

9. Find names and numbers of all salesperson who have more than one customer.

```
KD3-98843-Nandini> SELECT sname, snum FROM salespeople S
-> WHERE (SELECT COUNT(cnum) FROM customers C
-> WHERE C.snum = S.snum) > 1 ;
```

sname	snum
Peel	1001
Serres	1002

2 rows in set (0.02 sec)

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
KD3-98843-Nandini> SELECT S.snum, S.sname, S.city, COUNT(C.cnum) AS customer
_count FROM salespeople S
-> INNER JOIN customers C ON S.snum = C.snum GROUP BY S.snum, S.sname, S
.city HAVING COUNT(C.cnum) > 1;
```

snum	sname	city	customer_count
1001	Peel	London	2
1002	Serres	San Jose	2

2 rows in set (0.00 sec)

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
KD3-98843-Nandini> SELECT snum, SUM(amt) AS total_amount FROM orders
-> GROUP BY snum HAVING SUM(amt) >
-> (
-> SELECT MAX(amt)
-> FROM orders
-> );
```

snum	total_amount
1001	15382.07

1 row in set (0.00 sec)